

Haochen Sun

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University of Waterloo, 200 University Avenue West, Waterloo, Ontario, N2L 3G1, Canada

EDUCATION

University of Waterloo

PhD in Computer Science

- Supervisor: Prof. Xi He
- Research Focus: Security and privacy in machine learning and data management
- Honors and Awards: David R. Cheriton Graduate Scholarship

The Hong Kong University of Science and Technology (HKUST)

BSc in Data Science and Technology, and in Computer Science

- GPA: 3.888/4.3 | Major GPA: 4.041/4.3
- Honors and Awards: Chern Class Scholarship (Department of Mathematics), Zhiyuan Scholarship (China Soong Ching Ling Foundation), HKUST Scholarship Scheme for Continuing Undergraduates, Dean's List

Sep 2022 – Present
Waterloo, Canada

Sep 2018 – Jul 2022
Hong Kong, China

RESEARCH EXPERIENCE

Faithfulness of Differential Privacy Mechanism Execution

PhD Research Project (with Prof. Xi He)

- Investigating vulnerabilities from unfaithful execution of differential privacy mechanisms, focusing on additional privacy leakage incurred, and defenses to ensure execution integrity.

Zero-Knowledge Deep Learning

with Prof. Hongyang Zhang

- Specialized zero-knowledge proof (ZKP) protocols for deep learning with CUDA implementations.
- The first working ZKP scheme for 13B-parameter LLMs and for training 10M-parameter neural networks.

Air Pollution Forecasting with Deep Learning

Undergraduate Research and RA (with Profs. Jimmy Fung and Xingcheng Lu)

- Designed deep learning architectures for modeling the spatio-temporal distribution of air pollutants.
- Improved regional air pollution forecasting accuracy by 30%.

May 2024 – Present
University of Waterloo

Sep 2022 – Apr 2024
University of Waterloo

Jan 2020 – Nov 2022
HKUST

PUBLICATIONS

1. **(SIGMOD 2026 Best Paper Award)** [Haochen Sun](#) and Xi He. “GPM: The Gaussian Pancake Mechanism for Planting Undetectable Backdoors in Differential Privacy.” *Proceedings of the ACM on Management of Data (PACMMOD)*, 2026.
2. [Haochen Sun](#) and Xi He. “VDDP: Verifiable Distributed Differential Privacy under the Client–Server–Verifier Setup.” *Theory and Practice of Differential Privacy (TPDP)*, 2026.
3. [Haochen Sun](#)^{*}, Shufan Zhang^{*}, Karl Knopf^{*}, Shubhankar Mohapatra, Wei Pang, Calvin Wang, Yingke Wang, Masoumeh Shafieinejad, David Emerson, and Xi He. “FedDPSyn: Federated Tabular Data Synthesis with Computational Differential Privacy.” *Theory and Practice of Differential Privacy (TPDP)*, 2025.
4. [Haochen Sun](#), Jason Li, and Hongyang Zhang. “zkLLM: Zero-Knowledge Proofs for Large Language Models.” *ACM Conference on Computer and Communications Security (CCS)*, 2024.
5. [Haochen Sun](#), Tonghe Bai, Jason Li, and Hongyang Zhang. “zkDL: Efficient Zero-Knowledge Proofs of Deep Learning Training.” *IEEE Transactions on Information Forensics and Security (TIFS)*, 2024.
6. [Haochen Sun](#), Jimmy C. H. Fung, Yiang Chen, Zhenning Li, Dehao Yuan, Wanying Chen, and Xingcheng Lu. “Development of an LSTM broadcasting deep-learning framework for regional air pollution forecast improvement.” *Geoscientific Model Development (GMD)*, 2022.
7. [Haochen Sun](#), Jimmy C. H. Fung, Yiang Chen, Wanying Chen, Zhenning Li, Yeqi Huang, Changqing Lin, Mingyun Hu, and Xingcheng Lu. “Improvement of PM_{2.5} and O₃ forecasting by integration of 3D numerical simulation with deep learning techniques.” *Sustainable Cities and Society (SCS)*, 2021.

ACADEMIC SERVICES

(Sub)Reviewer: Conferences: CCS, SIGMOD, VLDB, NeurIPS, SaTML, AISTATS; Journals: TDSC, TMLR, TKDE.

Teaching Assistant, University of Waterloo: Courses: CS 116, CS 135, CS 246, CS 330, CS 480.